

R13 Physics

Complete Accurate Revision Guide

Waves & Light - Further Forces - Big & Small - Space

$$v = f\lambda$$

$$W = mg$$

$$P = F/A$$

$$\text{Moment} = Fd$$

$$F = kx$$

Detailed explanations, corrected misconceptions and labelled diagrams.

Focused on accuracy and complete exam content, not question padding.

Built for fast exam revision and accurate physics wording

Tip: learn the formula boxes, exact definitions and common mistakes pages first.

How to use this guide

This guide is designed to be a full revision document rather than a question pack. It includes your specification points, corrected scientific wording, diagrams and the connected facts that often appear in R13 end-of-year Physics exams.

Priority order for memorising: 1) formulas and units, 2) exact definitions, 3) diagrams and explanations, 4) common mistakes.

Topic	What you must be able to do
Waves	Explain energy transfer, oscillations, transverse/longitudinal waves, wave measurements, superposition and calculations.
Light	Explain reflection, refraction, dispersion, colour, filters, absorption/transmission/reflection and eye basics.
Further forces	Calculate/resultant forces, weight, moments, pressure and Hooke's law; explain balanced forces, elasticity, upthrust and gravity.
Big and small	Use quantities, symbols, units, prefixes and standard form correctly.
Space	Know solar system structure, day/night, seasons, universe expansion and Big Bang evidence.

Accuracy corrections: atmospheric pressure decreases with altitude. Two troughs overlapping add their negative displacements; destructive interference happens when a peak overlaps a trough. Light speed depends on optical density, not simply everyday material density.

Topic 1 - Waves

1.1 What waves do

A wave is a disturbance that transfers energy from one place to another. In most waves, particles in the medium oscillate around a fixed position. The particles do not travel along with the wave overall.

Exam phrase: Waves transfer energy by oscillations, without transferring matter overall.

Do not write: “waves transfer matter”. That is wrong. Particles may move backwards and forwards or up and down, but they remain around their equilibrium positions.

1.2 Oscillation and equilibrium

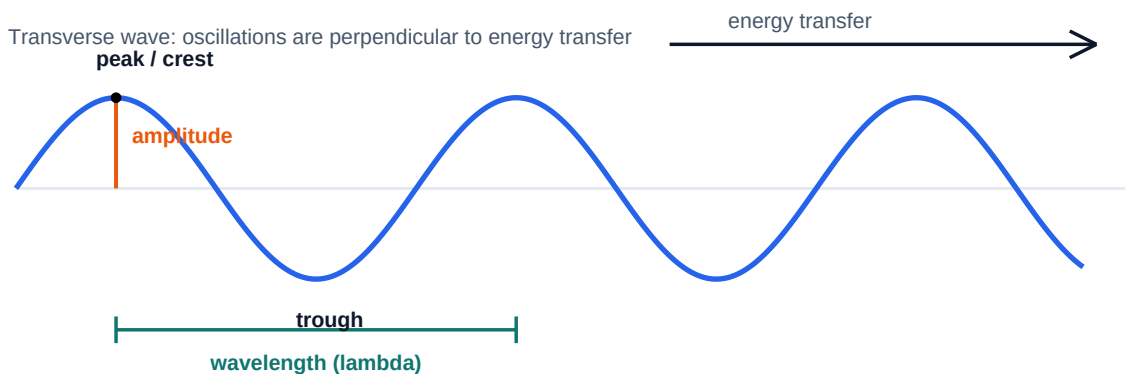
An **oscillation** is a repeated movement around a central/rest/equilibrium position. The equilibrium position is where the particle would be if no wave were passing.

- A vibrating guitar string oscillates around its resting position.
- A water particle in a wave moves around its position; it does not travel across the entire water surface with the wave.
- In sound, air particles oscillate backwards and forwards as the sound wave travels.

1.3 Transverse waves

In a **transverse wave**, the oscillations are perpendicular to the direction of energy transfer. “Perpendicular” means at 90 degrees.

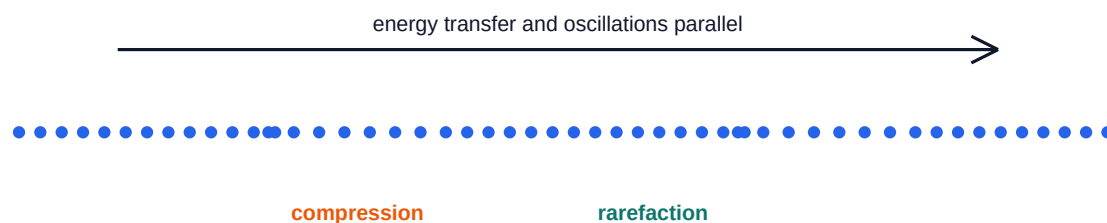
- Examples: light waves, electromagnetic waves, waves on a rope, water surface waves.
- Parts of a transverse wave include peaks/crests and troughs.
- Light is transverse and can travel through a vacuum because it is electromagnetic.



1.4 Longitudinal waves

In a **longitudinal wave**, the oscillations are parallel to the direction of energy transfer.

- Example: sound waves in air.
- Longitudinal waves have compressions and rarefactions.
- **Compression:** particles are closer together.
- **Rarefaction:** particles are further apart.



Longitudinal wave: particles vibrate backwards and forwards parallel to wave travel

1.5 Wave measurements

Measurement	Definition	Symbol / unit
Amplitude	Maximum displacement from the equilibrium/rest position.	metres (m)
Wavelength	Distance between two consecutive corresponding points, such as peak to peak or trough to trough.	lambda, metres (m)
Frequency	Number of complete waves passing a point each second.	f, hertz (Hz)
Period	Time taken for one complete oscillation/wave.	T, seconds (s)
Wave speed	Distance travelled by the wave per second.	v, metres per second (m/s)

$$\text{Wave speed} = \text{frequency} \times \text{wavelength} \quad v = f\lambda$$

$$\text{Period} = 1 \div \text{frequency} \quad T = 1/f$$

Example: if $f = 8 \text{ Hz}$ and $\lambda = 2.5 \text{ m}$, then $v = 8 \times 2.5 = 20 \text{ m/s}$. The answer is in m/s, not Hz/m.

1.6 Superposition and overlapping waves

When two waves meet, their displacements combine. This is called **superposition**. The final displacement at a point is the sum of the individual displacements.

Situation	Result	Accurate wording
Peak overlaps peak	Larger positive displacement	Constructive interference - displacements add.
Trough overlaps trough	Larger negative displacement	Constructive in size - negative displacements add, giving a deeper trough.
Peak overlaps trough	Smaller displacement or cancellation	Destructive interference - displacements partly or fully cancel.

Your sheet said “two troughs overlap their displacements are subtracted”. Strictly, two troughs have negative displacement, so they add to make a larger negative displacement. Peak + trough is the classic destructive case.

Waves - must-memorise sentences

- Waves transfer energy, not matter overall.
- Transverse waves have oscillations perpendicular to the direction of energy transfer.
- Longitudinal waves have oscillations parallel to the direction of energy transfer.
- Frequency is measured in Hz and means complete waves per second.
- Amplitude is maximum displacement from equilibrium position.
- Wavelength is the distance between consecutive corresponding points.

Topic 2 - Light

2.1 What light is

Light is a transverse electromagnetic wave. It can travel through a vacuum at approximately **300,000,000 m/s**, written in standard form as **$3.0 \times 10^8 \text{ m/s}$** .

- Light travels in straight lines through a uniform medium.
- Light can be reflected, transmitted, absorbed or refracted.
- Light travels more slowly in optically denser materials such as glass than in air or a vacuum.

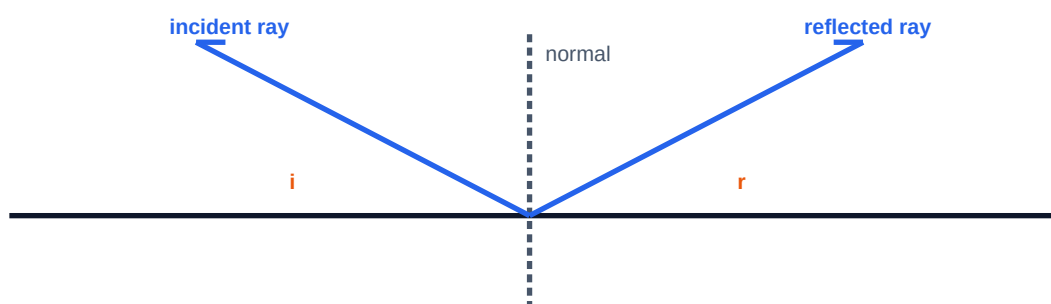
Optically denser means light travels more slowly in that medium. This is not exactly the same thing as ordinary mass density.

2.2 Reflection

Reflection happens when light bounces off a surface. The law of reflection is:

Angle of incidence = angle of reflection

Both angles are measured from the **normal**, not from the mirror surface. The normal is an imaginary line drawn at 90 degrees to the surface.



Law of reflection: angle i = angle r , both measured from the normal

2.3 Specular and diffuse reflection

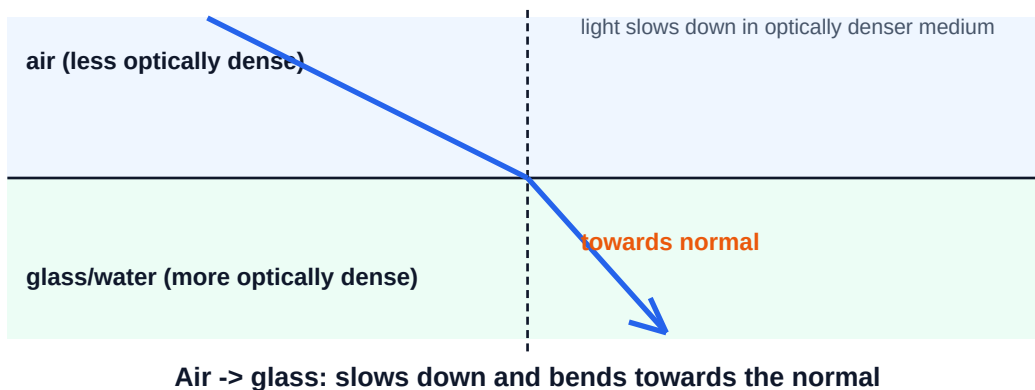
Type	Surface	What happens to rays	Image?
Specular reflection	Smooth, flat surface	Parallel incident rays reflect and remain organised/parallel.	Clear image formed
Diffuse reflection	Rough surface	Rays reflect in many different directions because the surface has many tiny angles.	No clear image formed

Exam phrase: diffuse reflection scatters reflected rays in different directions, so no clear image is formed.

2.4 Refraction

Refraction occurs when light changes speed as it passes from one medium into another. If the light enters at an angle, the change in speed causes it to change direction.

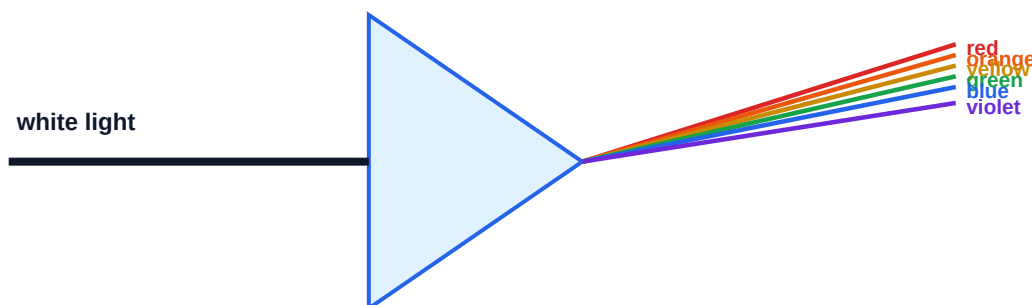
Light path	Speed change	Direction change
Less optically dense $\rightarrow$ more optically dense, e.g. air to glass	Slows down	Bends towards the normal
More optically dense $\rightarrow$ less optically dense, e.g. glass to air	Speeds up	Bends away from the normal
Enters along the normal (90 degrees to surface)	Speed changes	Does not bend, because it is not entering at an angle



2.5 Dispersion

White light is a mixture of all the colours of the visible spectrum. When white light passes through a triangular prism, the colours separate because each colour refracts by a different amount.

- Red has the lowest frequency of visible light and bends the least.
- Violet has the highest frequency of visible light and bends the most.
- The order of colours is red, orange, yellow, green, blue, indigo, violet.



Dispersion: colours refract by different amounts; violet bends most, red least

Do not just say “the prism breaks the light”. Say: white light contains many colours, and different colours refract by different amounts.

2.6 Colour and filters

Objects appear coloured because they reflect some colours of light and absorb the others. Filters transmit some colours and absorb others.

Situation	What happens	Appearance
Red object in white light	Reflects red, absorbs most other colours	Red
Blue object under red light	Blue object cannot reflect blue because no blue light is present	Black/dark
Green object through red filter	Green light from object is absorbed by red filter; red filter transmits red only	Black/dark
White object under blue light	Reflects the blue light available	Blue
Black object in white light	Absorbs most light, reflects very little	Black

2.7 The human eye

Part	Function
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Cornea	Transparent curved front surface; refracts most incoming light.
Pupil	Opening that allows light into the eye.
Iris	Coloured muscle that changes pupil size to control light entering.
Lens	Focuses light onto the retina by changing shape.
Ciliary muscles	Change the shape of the lens for focusing/accommodation.
Retina	Light-sensitive layer containing receptor cells; image forms here.
Optic nerve	Carries electrical impulses from the retina to the brain.

Accommodation: the eye changes lens shape to focus on near or distant objects. Near objects require a thicker, more curved lens; distant objects require a thinner lens.

Topic 3 - Resultant forces and motion

3.1 What a force is

A force is a push or pull. Forces are vector quantities, meaning they have size and direction. Force is measured in newtons, N.

3.2 Resultant force

The resultant force is a single force that has the same effect as all the forces acting on an object combined.

Forces acting	How to find resultant	Example
Same direction	Add the forces	8 N right + 5 N right = 13 N right
Opposite directions	Subtract smaller from larger; direction is the larger force direction	20 N left - 12 N right = 8 N left
Balanced	Forces cancel, resultant = 0 N	10 N up and 10 N down = 0 N

3.3 Resultant force and motion

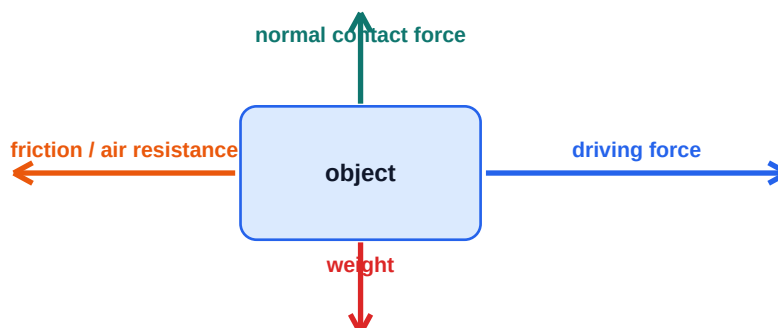
Resultant force	Motion
0 N	Object remains stationary OR continues moving at constant speed in a straight line.
Not 0 N	Object accelerates in the direction of the resultant force.
Acceleration	Can mean speeding up, slowing down, or changing direction.

If the resultant force is zero, it does not mean the object must be stationary. It can also move at constant velocity.

3.4 Free-body diagrams

A free-body diagram shows all the forces acting on one object. Forces are shown with vector arrows.

- Arrow direction shows direction of force.
- Arrow length can show relative size of force.
- Each arrow should be labelled, e.g. weight, friction, thrust, air resistance, tension, normal contact force.
- Only draw forces acting on the object, not forces the object exerts on other things.



Free-body diagrams show forces on one object only. Arrow length shows relative size.

Topic 4 - Weight, mass and gravity

4.1 Mass and weight

Term	Meaning	Unit
Mass	Amount of matter in an object. It does not change with location.	kilogram (kg)
Weight	Force on an object due to gravity. It changes if gravitational field strength changes.	newton (N)
Gravitational field strength	Force per kilogram on a mass in a gravitational field.	N/kg

$$\text{Weight} = \text{mass} \times \text{gravitational field strength} \quad W = mg$$

On Earth, gravitational field strength is about **9.8 N/kg**. Many school questions use **10 N/kg** for easier calculations.

4.2 Gravity

A gravitational force acts between any two objects with mass. It is always attractive.

- The bigger the masses, the bigger the gravitational force.
- The greater the distance between the masses, the smaller the gravitational force.
- Near Earth, objects fall with acceleration about 9.8 m/s^2 if air resistance is ignored. Your school may round this to 10 m/s^2 .

Weight is not measured in kg. Weight is a force and is measured in N. Mass is measured in kg.

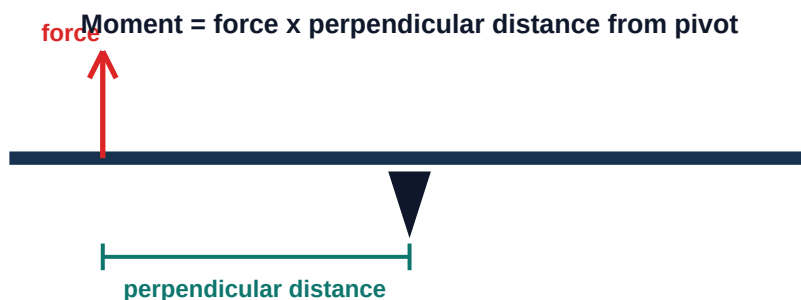
Topic 5 - Moments

5.1 Moment meaning

A moment is the turning effect of a force around a pivot. Moments explain levers, seesaws, doors, spanners and balances.

$$\text{Moment} = \text{force} \times \text{perpendicular distance from pivot}$$

$$M = Fd \quad \text{unit: newton metre (Nm)}$$



5.2 Perpendicular distance

The distance in the moments equation is not always the length of the object. It is the **perpendicular distance** from the pivot to the line of action of the force.

A door handle is far from the hinge to increase the perpendicular distance, increasing the moment for the same force.

5.3 Principle of moments

For an object to be balanced and not rotate:

$$\text{Total clockwise moment} = \text{total anticlockwise moment}$$

Example: 20 N acting 2 m from a pivot gives 40 Nm. To balance it, 10 N must act 4 m from the pivot because $10 \times 4 = 40$ Nm.

Topic 6 - Forces and elasticity

6.1 Elastic and plastic deformation

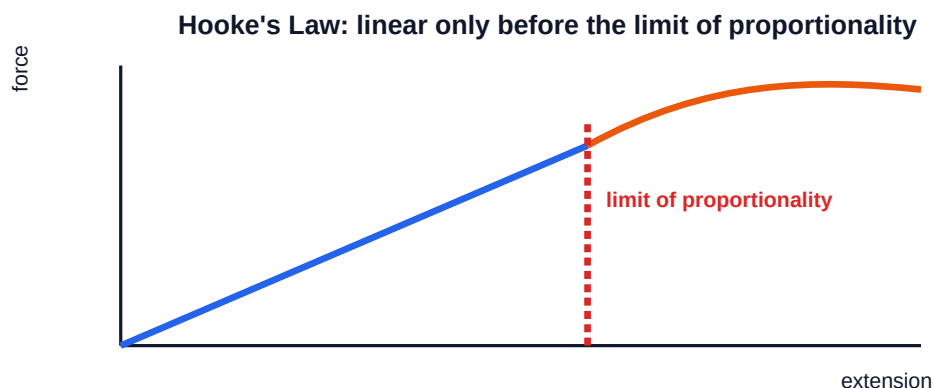
Term	Definition
Elastic object	An object that returns to its original shape after being stretched or compressed.
Elastic deformation	Temporary change in shape; object returns to original length/shape when force removed.
Plastic deformation	Permanent change in shape; object does not fully return.
Elastic potential energy	Energy stored when an elastic object is stretched or compressed.

6.2 Hooke's Law

If an elastic object obeys Hooke's law, the extension is directly proportional to the force applied, as long as the limit of proportionality has not been exceeded.

$$\text{Force} = \text{spring constant} \times \text{extension} \quad F = kx$$

Symbol	Meaning	Unit
F	Force applied	N
k	Spring constant/stiffness	N/m
x	Extension = stretched length - original length	m



6.3 Spring constant and stiffness

The spring constant measures how stiff a spring is. A large spring constant means a larger force is needed for the same extension.

Example: $k = 80 \text{ N/m}$ and $x = 0.15 \text{ m}$ gives $F = 80 \times 0.15 = 12 \text{ N}$.

6.4 Factors affecting how easy something is to stretch

- Material: rubber stretches more easily than steel for the same shape.
- Length: longer objects often stretch more for the same force.
- Thickness/cross-sectional area: thicker objects are usually harder to stretch.
- Temperature: higher temperature can make some materials easier to stretch.
- Shape/structure: coils, fibres and woven structures affect stiffness.

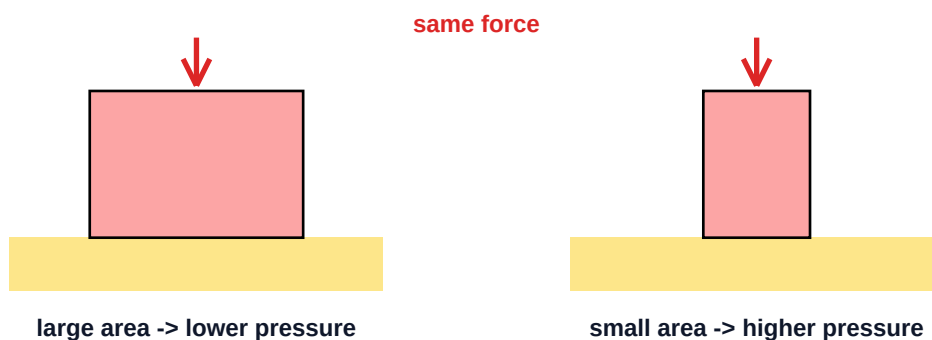
Topic 7 - Pressure and fluids

7.1 What pressure is

Pressure is force per unit area. It tells you how concentrated a force is over a surface.

$$\text{Pressure} = \text{force} \div \text{area} \quad P = F/A$$

Quantity	Symbol	Unit
Pressure	P	pascal (Pa)
Force	F	newton (N)
Area	A	metre squared (m ²)



7.2 Area and pressure

- If force increases and area stays the same, pressure increases.
- If area increases and force stays the same, pressure decreases.
- Sharp knives cut better because the same force acts over a smaller area, producing greater pressure.
- Snowshoes reduce pressure because the person's weight is spread over a larger area.

7.3 Atmospheric pressure

Atmospheric pressure is caused by gas particles in the air colliding with surfaces and by the weight of the air above us.

As altitude increases, atmospheric pressure decreases because there are fewer air particles above you and less weight of air pressing down.

Correct version: atmospheric pressure decreases with altitude. If your notes say increases, that is a mistake.

7.4 Upthrust and floating

Upthrust is an upward force exerted by a fluid on an object. It happens because pressure in a fluid increases with depth, so the pressure on the bottom of an object is greater than the pressure on the top.

- If upthrust equals weight, the object floats or remains suspended.
- If weight is greater than upthrust, the object sinks.
- A large steel ship can float because its overall density, including the air-filled space inside it, is less than the density of water.

Floating depends on overall density and upthrust, not simply on whether the material itself is "heavy".

Topic 8 - Big and small numbers, quantities and units

8.1 Quantities, symbols and units

A **quantity** is something that can be measured. A **unit** is the standard amount used to measure that quantity. Quantities are often represented by symbols; units are represented by abbreviations.

Quantity	Common symbol	Unit name	Unit abbreviation
Mass	m	kilogram	kg
Length/distance	d, s or l	metre	m
Time	t	second	s
Temperature	T	degrees Celsius / kelvin	degreesC / K
Energy	E	joule	J
Current	I	ampere/amp	A
Voltage / potential difference	V	volt	V
Resistance	R	ohm	Ω
Frequency	f	hertz	Hz
Force	F	newton	N
Pressure	P	pascal	Pa

8.2 Prefixes

Prefix	Symbol	Multiplier	Meaning
nano	n	10^{-9}	one billionth
micro	micro	10^{-6}	one millionth
milli	m	10^{-3}	one thousandth
centi	c	10^{-2}	one hundredth
kilo	k	10^3	one thousand
mega	M	10^6	one million
giga	G	10^9	one billion

When using equations, values must be in the correct SI units: kg, m, s, N, Pa, m/s, Hz, etc.

8.3 Conversion examples

Conversion	Method	Answer
4.8 km to m	$\times 1000$	4800 m
3500 g to kg	$\div 1000$	3.5 kg
250 mA to A	$\div 1000$	0.25 A
0.006 m to mm	$\times 1000$	6 mm
3.2 MHz to Hz	$\times 1,000,000$	3,200,000 Hz

8.4 Standard form

Standard form writes very large or small numbers as a number between 1 and 10 multiplied by a power of 10.

$a \times 10^n$ where $1 \leq a < 10$

Ordinary number	Standard form
300,000,000	3.0×10^8
72,000	7.2×10^4
0.00054	5.4×10^{-4}
0.000072	7.2×10^{-5}

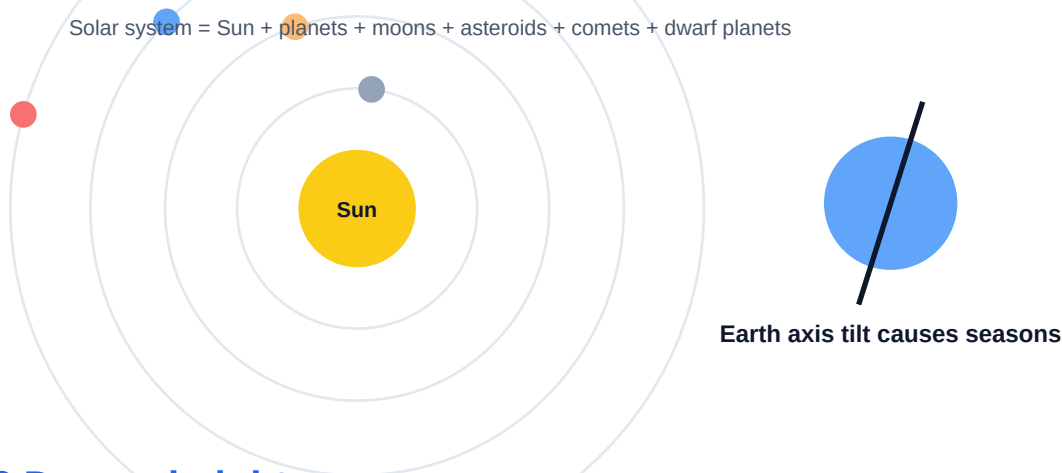
Do not write 54×10^{-5} as final standard form because 54 is not between 1 and 10. It should be 5.4×10^{-4} .

Topic 9 - Space

9.1 The solar system

The solar system is made of the Sun and the objects held in orbit around it by gravity.

- Sun: the star at the centre of the solar system.
- Planets: eight major planets orbiting the Sun.
- Moons: natural satellites orbiting planets.
- Asteroids: rocky objects, many found in the asteroid belt.
- Comets: icy bodies that can form tails when near the Sun.
- Dwarf planets: objects such as Pluto that orbit the Sun but are not classed as full planets.



9.2 Day and night

Day and night are caused by Earth rotating on its axis. One full rotation takes about 24 hours. The side facing the Sun experiences day, and the side facing away experiences night.

9.3 Seasons

Seasons are caused by the tilt of Earth's axis, not by Earth being closer to or further from the Sun.

- When the northern hemisphere is tilted towards the Sun, it gets more direct sunlight and longer days: summer.
- When the northern hemisphere is tilted away from the Sun, it gets less direct sunlight and shorter days: winter.
- The southern hemisphere has opposite seasons to the northern hemisphere.

Common mistake: "seasons happen because Earth is closer to the Sun in summer". That is wrong for school Physics. The cause is axial tilt.

9.4 The universe and Big Bang

The universe contains all space, time, matter and energy. It includes galaxies, stars, planets, gas, dust, radiation and dark matter/dark energy concepts at higher levels.

The Big Bang model states that the universe began around **13.8 billion years ago** in a very hot, dense state and has been expanding ever since.

Evidence	Meaning
Red shift	Light from distant galaxies is shifted towards longer wavelengths, showing galaxies are moving away and the universe is expanding.
Cosmic microwave background radiation	Leftover radiation from the early universe, now detected as microwave radiation in all directions.

Extra Accuracy Page 1 - Waves and light precision

Energy transfer and mediums

The safest exam wording is: **waves transfer energy without transferring matter overall**. In mechanical waves, particles of the medium oscillate around equilibrium positions. Electromagnetic waves, including light, do not need a material medium and can travel through a vacuum.

Wave type	Needs medium?	Oscillation / example
Mechanical	Yes	Particles oscillate: sound in air, water waves, waves on a rope.
Electromagnetic	No	Electric and magnetic fields oscillate: light, radio waves, microwaves.

Do not say all waves need a medium. Sound does, but light does not.

Superposition - corrected wording

When waves overlap, the displacement at a point is the sum of the individual displacements. Peaks are positive displacement; troughs are negative displacement.

Overlap	Accurate result
Peak + peak	Constructive: bigger peak. Displacements add.
Trough + trough	Constructive in size: deeper trough. Negative displacements add.
Peak + trough	Destructive: displacements reduce or cancel.

Your original sheet wording about two troughs being subtracted is scientifically imprecise. Peak + trough is the destructive case.

Refraction precision

Path	Speed	Direction
Air -> glass	Slows down	Bends towards the normal if entering at an angle.
Glass -> air	Speeds up	Bends away from the normal if entering at an angle.
Along the normal	Speed changes	No bending because it enters at 90 degrees.

Extra Accuracy Page 2 - Colour, filters and the eye

Colour rules

An object can only reflect colours that are present in the incoming light. A filter only transmits certain colours and absorbs others. If the colour needed cannot reach the eye, the object appears black or very dark.

Situation	Reason	Appearance
Blue object through red filter	Blue light is reflected by object but absorbed by red filter.	Black/dark
Red object in blue light	No red light available to reflect.	Black/dark
White object in green light	White object reflects the green light available.	Green
Black object in white light	Absorbs most visible light.	Black

Dispersion exact wording

White light contains all visible colours. A prism separates the colours because each colour refracts by a different amount. Violet has the highest frequency and bends the most; red has the lowest frequency and bends the least.

Do not write “the prism breaks light”. Write: different colours refract by different amounts.

Human eye - high-risk distinctions

Part	Function
Cornea	Refracts most incoming light.
Lens	Fine-focuses light onto the retina.
Retina	Contains receptor cells and detects the image.
Ciliary muscles	Change lens shape for accommodation.
Iris / pupil	Iris controls pupil size; pupil is the hole light enters through.

Retina = detects light. Lens = focuses light. Ciliary muscles = change lens shape.

Extra Accuracy Page 3 - Further forces precision

Free-body diagrams

A free-body diagram shows the forces acting on one object only. It does not show forces that the object exerts on other objects.

Force	Direction / situation
Weight	Downwards, towards centre of Earth.
Normal contact force	Perpendicular to the surface supporting the object.
Friction	Opposes sliding or attempted sliding.
Air resistance / drag	Opposes motion through air or fluid.
Tension	Along a rope, string or cable.
Thrust / driving force	Direction of propulsion or push.

Resultant force and motion

Resultant force	Motion meaning
0 N	Stationary or moving at constant velocity; no acceleration.
Non-zero	Object accelerates in the direction of the resultant force.
Opposite direction to motion	Object slows down/decelerates.
Sideways to motion	Object changes direction, so velocity changes.

A resultant force does not simply mean “it moves that way”. It means the object accelerates that way.

Mass, weight and gravity

Mass is the amount of matter and is measured in kg. Weight is the force due to gravity and is measured in N. Gravitational field strength on Earth is about 9.8 N/kg, often rounded to 10 N/kg.

$$W = mg$$

The bigger the masses, the stronger the gravitational force. The greater the distance, the weaker the gravitational force.

Extra Accuracy Page 4 - Elasticity, pressure, units and space

Hooke's Law and elasticity

Hooke's law applies only up to the limit of proportionality. Before this limit, force and extension are directly proportional, so a force-extension graph is a straight line through the origin.

$$F = kx$$

- Large spring constant = stiff spring = harder to stretch.
- Extension = stretched length - original length.
- Elastic deformation is temporary; plastic deformation is permanent.
- Elastic potential energy is stored when an elastic object is stretched or compressed.

Pressure and upthrust

Pressure is force per unit area. A smaller area gives a larger pressure for the same force. Pressure in liquids increases with depth, so the bottom of an object experiences greater pressure than the top. This pressure difference creates upthrust.

$$P = F/A$$

Upthrust compared with weight	Result
Upthrust > weight	Object accelerates upwards.
Upthrust = weight	Object floats or stays at constant depth.
Upthrust < weight	Object sinks.

Atmospheric pressure decreases with altitude because there are fewer air particles above you and less weight of air pressing down.

Units, standard form and space wording

- Always convert to equation units: kg, m, s, N, Pa, Hz and m/s.
- micro = 10^{-6} , milli = 10^{-3} , centi = 10^{-2} , kilo = 10^3 , mega = 10^6 .
- Standard form must be $a \times 10^n$ where $1 \leq a < 10$.
- Day and night are caused by Earth rotating. Seasons are caused by Earth's tilted axis, not distance from the Sun.
- The Big Bang was the expansion of space itself; evidence includes red shift and cosmic microwave background radiation.

Final must-memorise checklist

Core formulas

Formula	Used for	Units
$v = \lambda f$	Wave speed	v m/s, f Hz, λ m
$W = mg$	Weight	W N, m kg, g N/kg
$P = F/A$	Pressure	P Pa, F N, A m ²
Moment = Fd	Turning effect	Moment Nm, F N, d m
$F = kx$	Hooke's law	F N, k N/m, x m
$T = 1/f$	Wave period	T s, f Hz

Definitions you should know word-for-word

- **Resultant force:** single force with the same effect as all the forces acting on an object.
- **Weight:** force on an object due to gravity.
- **Mass:** amount of matter in an object.
- **Moment:** turning effect of a force.
- **Pressure:** force per unit area.
- **Elastic object:** returns to its original shape after being stretched or compressed.
- **Amplitude:** maximum displacement from equilibrium position.
- **Frequency:** number of complete waves per second.
- **Wavelength:** distance between consecutive corresponding points on a wave.
- **Refraction:** change in direction caused by a wave changing speed when entering a different medium.

High-risk mistakes

Wrong	Correct
Waves transfer matter.	Waves transfer energy, not matter overall.
Angles in reflection measured from mirror surface.	Angles are measured from the normal.
Sound is transverse.	Sound is longitudinal.
Weight is measured in kg.	Weight is measured in N. Mass is measured in kg.
Atmospheric pressure increases with altitude.	Atmospheric pressure decreases with altitude.
Red bends most in a prism.	Violet bends most; red bends least.
Zero resultant force means stationary only.	It can mean stationary OR constant velocity.
Hooke's law always applies.	Only applies up to the limit of proportionality.

One-hour emergency revision order

- First 10 minutes: formulas and units.
- Next 15 minutes: waves and light definitions.
- Next 15 minutes: forces, moments, Hooke's law and pressure.
- Next 10 minutes: prefixes, conversions and standard form.
- Final 10 minutes: common mistakes and diagrams.